

Introduction to Artificial Intelligence

Quantify Uncertainty

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23/04/2026

Plan

So far:

- Defined agents
- Environment with perfect information and deterministic environments
- Factoring the environment

Today:

- Reasoning under uncertainty
- Recall basic probability theory

Acting Under Uncertainty

Uncertainty

More often than not, systems need to handle uncertainty because of different reasons:

- Partially observable environment (e.g., Poker game)
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- Imprecise sensors and actuators

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How can a system make decisions under these conditions?

Perspectives on Uncertainty

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Interlude: Fuzzy Logic

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Dealing with vague or imprecise information:

- Fuzzy sets are **non-probabilistic** representations of uncertainty
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- **Syntax:** same as Propositional Logic
- **Semantics:** every sentence maps to a real number in $[0, 1]$:

Sentence	Value
$\top$	1
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$P \vee Q$	$\max([P], [Q])$
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Fuzzy logic can be used as a formalism to **reason under uncertainty**.

Fuzzy Logic: Applications

- Widely used in control systems
- Decision making under uncertainty
- Modelling natural language semantics

End of the interlude

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Decision theory = probability theory + utility theory

Probability Theory

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- Sample space: set of all possible 'atomic' outcomes
- Events: *relevant* subsets of the sample space
- Probability function: assigns a measure to each event

Basic Probability Notions

Definition: Probability Function

A **probability space** $\langle \Omega, \Sigma, P \rangle$ where

- Ω is a sample space
- Σ is the set of events
- $P : \Sigma \rightarrow \mathbb{R}$ is a probability function assigning a probability to each possible event $E \in \Sigma$ such that:

Ax1: $P(E) \geq 0$ for all $E \in \Sigma$

Ax2: $P(\Sigma) = 1$

Ax3: $P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$ for any $E_1, E_2, \dots$ of mutually disjoint

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- $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

Examples

Example: Coin Toss

$$\Omega = \{H, T\}$$

$$\Sigma = \mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \Omega\}$$

$$\begin{array}{rcl} P : & \Sigma & \mapsto [0, 1] \\ & E & \mapsto P(E) \end{array}$$

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- $E = \{\text{at least one head}\} = \{HH, HT, TH\} \implies P(E) = 0.75$
- $E = \{\text{exactly one head}\} = \{HT, TH\} \implies P(E) = 0.50$

Examples

Example: Two Dices

$$\Omega = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$$

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Events:

- $E = \{\text{at least one six}\} = \{(6, i) \mid i \in \{1, \dots, 6\}\}$
 $P(E) = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = 6 \cdot \frac{1}{36} = \frac{1}{6}$

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- $E = \{\text{exactly one six}\} = \{(i, 6) \mid i \in \{1, \dots, 5\}\}$
 $P(E) = 5 \cdot \frac{1}{36} = \frac{5}{36}$

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$P = \text{the measure of the area of the region}$

Intuitively:

- the probability of hitting a specific point on the dartboard is zero (because it has no area)
- the probability of hitting a region (e.g., the bullseye) can be non-zero.

Bayesian Reasoning

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Definition: Conditional Probability

The probability of E given the **evidence** F :

- **product rule:**

$$P(E \wedge F) = P(E \mid F) \cdot P(F)$$

- **quotient rule:**

$$P(E \mid F) = \frac{P(E \wedge F)}{P(F)} \quad \text{if} \quad P(F) > 0$$

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Q: What can we say if the data is the following?

- $\frac{6}{10}$ of students attend the TA sessions *and* pass the exam;
- $\frac{1}{4}$ of students attend the TA sessions.

Bayes' Rule

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Theorem: Bayes' Rule

If we have evidence E and want to know the probability of a conclusion C , we can use Bayes' rule:

$$P(C | E) = \frac{P(C) \cdot P(E | C)}{P(E)}$$

- $P(C)$ — **prior** probability of the conclusion
- $P(E | C)$ — **likelihood** of the evidence given the conclusion
- $P(E)$ — **marginal** probability of the evidence
- $P(C | E)$ — **posterior** probability of the conclusion given evidence

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Bayes' Rule – Example

Free candy:

- 1% of customers are secretly elected as pro customers ($P(\text{Pro})$)
- pro customers get a free candy 80% of the time ($P(\text{Candy} \mid \text{Pro})$)
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$$P(\text{Candy}) = P(\text{Candy} \mid \text{Pro}) \cdot P(\text{Pro}) + P(\text{Candy} \mid \neg\text{Pro}) \cdot P(\neg\text{Pro})$$

Bayesian Reasoning

Random Variables

A **random variable** is a function $X: \Omega \mapsto S$ from a sample space Ω to a set of possible values S .

Example:

A random variable for the weather (W), which can take on values in $S = \{\text{sun, rain, cloud, snow}\}$.

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We then associate probabilities to each possible value of the random variable:

$$\begin{array}{ll} P(W = \text{sun}) &= 0.60 \\ P(W = \text{cloud}) &= 0.29 \end{array} \qquad \begin{array}{ll} P(W = \text{rain}) &= 0.10 \\ P(W = \text{snow}) &= 0.01 \end{array}$$

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The function P defines the ***probability distribution*** of the random variable Weather.

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Given two (or more) random variables, we can define their joint distribution $P(X, Y)$ which is a probability distribution over the Cartesian product of their possible values.

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Random variables X and Y are **independent** if $P(X \wedge Y) = P(X) \cdot P(Y)$ or equivalently if $P(X \mid Y) = P(X)$.

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$$P(X \mid Y = y) = \frac{P(X, Y = y)}{P(Y = y)}$$

Chain rule = expressing the joint distribution of a set of random variables in terms of their conditional distributions.

$$P(X_1, \dots, X_n) = P(X_1 \mid X_2, \dots, X_n) \cdot P(X_2 \mid X_3, \dots, X_n) \cdots P(X_n)$$

Summary

Book references:

- Chapter 12 (see also Appendix A)

Next lectures:

- Bayesian Networks
- Markov Decision Processes